

# Teacher Guide

PowerSat CubeSat Mission  
STEMSat Educational CubeSat Program

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## 1. Mission Overview

In this activity, students investigate how energy is generated, managed, and used within a spacecraft system. Students build and operate a CubeSat-inspired device that simulates a satellite electrical power subsystem (EPS) using a light sensor module to represent solar energy input.

Students conduct experiments by varying light conditions to simulate solar energy input and observe how limited energy affects system performance. Data is transmitted to a computer through the Serial Monitor or Bluetooth module, simulating how real satellites send telemetry to a ground station.

Through testing and data analysis, students explore the relationship between energy input, power demand, and system behavior while applying the engineering design process to evaluate and improve their CubeSat power system.

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## 2. Target Grade Level

Grades: 6–8 (Middle School)

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## 3. Time Requirements

| Activity             | Time          |
|----------------------|---------------|
| Mission introduction | 10 minutes    |
| Hardware assembly    | 10–15 minutes |

|                                 |               |
|---------------------------------|---------------|
| Code upload and verification    | 10 minutes    |
| Energy experiments              | 20–25 minutes |
| Data analysis and discussion    | 15 minutes    |
| Engineering redesign discussion | 10 minutes    |

**Total Time: ~75–85 minutes**

This activity can also be divided into two class periods.

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## 4. Learning Objectives

By the end of the activity, students will be able to:

1. Explain how satellites generate energy using solar power
  2. Investigate how light intensity affects energy generation
  3. Interpret telemetry data to determine system behavior
  4. Understand why energy is a limiting factor in spacecraft design
  5. Apply engineering design principles to improve a system
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## 5. NGSS Alignment

### Engineering Design

| Standard  | Application                                                   |
|-----------|---------------------------------------------------------------|
| MS-ETS1-1 | Students define the problem of limited energy in a system     |
| MS-ETS1-2 | Students evaluate how different conditions affect performance |
| MS-ETS1-3 | Students analyze data to identify patterns and improve design |
| MS-ETS1-4 | Students test and refine their CubeSat energy system          |

### Science Standards

| Standard | Application                                                              |
|----------|--------------------------------------------------------------------------|
| MS-PS3-3 | Students apply scientific principles to design and test an energy system |

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## 6. Materials

### Per Student Group (2–4 students)

- Arduino Uno
- Light sensor module
- HC-05 Bluetooth module
- Female to female jumper wires
- Cable holders (optional)
- USB cable
- Computer with Arduino IDE
- CubeSat frame

### Instructor Materials

- Pre-installed Arduino libraries
  - Pre-tested sensors
  - Flashlights or adjustable light sources
  - Printed mission guides
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## 7. Pre-Class Setup (Instructor)

### Install Arduino IDE

Install on all computers.

Download:

<https://www.arduino.cc/en/software>

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### Install Required Libraries

PowerSat uses minimal libraries.

- SoftwareSerial (only if Bluetooth is used)

How to install/check:

1. Open Arduino IDE
  2. Go to Tools → Manage Libraries
  3. Search: SoftwareSerial
  4. Install if not already installed
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## **Test Components / Make Example Model**

Before class:

1. (Optional but highly recommended) Pre-attach wiring to sensors (refer to wiring instructions in the mission guide) and optionally use a 3D printed connector holder to hold wires in place and keep students from mixing up wiring for sensors when connecting to the Arduino.
  2. Follow the wiring instructions in the Mission Guide to attach sensors and upload code to one Arduino to ensure everything is working/to create an example model for students
  3. Verify readings appear in the serial monitor
  4. Light readings change when the sensor is covered/uncovered, and the system state (ON / DIM / OFF) changes with light level
  5. Pre-check that the Bluetooth module can connect and print out values wirelessly to the serial monitor
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## **Prepare Test Stations**

Set up areas for experiments, such as:

- Flashlight testing station
  - Low-light or shaded area
  - Covered sensor setup (simulate eclipse)
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# **8. Running the Activity**

## **Step 1: Introduce the Mission**

Ask students:

- How do satellites get energy in space?

- What happens when a satellite is in darkness?
- Why might systems need to turn off?

Explain that students will simulate a satellite power system using light as the energy source.

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## Step 2: Assemble the CubeSat

- Students connect the light sensor
  - Instructor checks wiring before power is applied.
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## Step 3: Upload the Code

Students upload the mission code and observe output.

Confirm that:

- Light increases → system state becomes ON
  - Light decreases → system becomes DIM or OFF
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# 9. Energy Experiments

Students perform three energy experiments.

| Test              | Purpose                        |
|-------------------|--------------------------------|
| Bright Light Test | Simulate maximum solar input   |
| Dim Light Test    | Simulate reduced energy        |
| Eclipse Test      | Simulate no sunlight           |
| High Demand Test  | Simulate increased power usage |

Students record system state during each test.

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## 10. Expected Results

Typical observations:

| Condition      | Observation |
|----------------|-------------|
| Bright light   | System ON   |
| Moderate light | System DIM  |
| Low/no light   | System OFF  |

Students should observe that reduced energy leads to reduced system performance.

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## 11. Data Analysis

Students analyze:

- Which condition produced the highest light level
- Which condition caused system shutdown
- How power demand affected system state

Instructor discussion prompts:

- Why does reduced light affect system behavior?
  - What happens to satellites during eclipse?
  - Why is energy management critical in space?
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## 12. Engineering Design Discussion

Students evaluate the CubeSat design.

Possible design improvements:

- Reduce power demand
- Improve efficiency of system operation
- Adjust thresholds for system response
- Optimize sensor placement

This reinforces the engineering design cycle.

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## 13. Troubleshooting Guide

### **Problem: Light readings do not change**

Possible cause:

- Incorrect wiring

Fix:

- Verify A0 connections

### **Problem: System state does not change**

Possible cause:

- Threshold values too high/low

Fix:

- Adjust threshold values in code

### **Problem: No data output**

Possible cause:

- Serial/Bluetooth not connected

Fix:

- Check COM port and connections
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## 14. Safety Considerations

- Avoid shining bright lights directly into eyes
- Keep wires secured during operation

- Handle electronics carefully
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## **15. Extension Activities**

### **Extension 1: Power Prioritization**

Students decide how a system should respond when energy is limited

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### **Extension 2: Day/Night Cycle Simulation**

Students alternate between light and dark conditions

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### **Extension 3: System Optimization**

Students modify thresholds or design to improve performance

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## **16. Real-World Space Connection**

Satellites rely on solar panels and onboard batteries to operate. When satellites enter Earth's shadow, they must rely on stored energy and carefully manage which systems remain active.

Similar systems are used in:

- Earth observation satellite
- Communication satellites
- Deep space probes

The PowerSat mission allows students to simulate real spacecraft power challenges.

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## **17. Mission Success Criteria**

The mission is successful if:

- CubeSat responds to changes in light conditions

- Students perform multiple energy experiments
- Students interpret telemetry data
- Students propose improvements to the CubeSat design